

Algebra II with Trigonometry

Chapter 11.3

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4666324_alg-2trig-h-chp-11-3-note-docx.pdf](#)

Solutions

1. A hyperbola centered at the origin has asymptotes $y = \pm 2x$ and passes through $(3, \sqrt{13})$. Find its standard equation.

Since the asymptotes are $y = \pm 2x$, the hyperbola must be of one of the forms

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{or} \quad \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

with asymptote slopes determined by $\pm \frac{b}{a}$ or $\pm \frac{a}{b}$.

Because the point $(3, \sqrt{13})$ has $x^2 = 9$ and $y^2 = 13$, we note that $y^2 > x^2$, so it is natural to expect the transverse axis to be vertical. Then the equation is

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

whose asymptotes are

$$y = \pm \frac{a}{b}x.$$

Given slope 2, we have

$$\frac{a}{b} = 2 \Rightarrow a = 2b.$$

Now substitute $(3, \sqrt{13})$:

$$\frac{13}{a^2} - \frac{9}{b^2} = 1.$$

Using $a^2 = 4b^2$,

$$\frac{13}{4b^2} - \frac{9}{b^2} = 1 \Rightarrow \frac{13 - 36}{4b^2} = 1 \Rightarrow -\frac{23}{4b^2} = 1,$$

impossible. So the hyperbola is not vertical.

Hence it must be

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

whose asymptotes are

$$y = \pm \frac{b}{a}x.$$

Given slope 2, we get

$$\frac{b}{a} = 2 \Rightarrow b = 2a.$$

Substitute $(3, \sqrt{13})$:

$$\frac{9}{a^2} - \frac{13}{b^2} = 1.$$

Since $b^2 = 4a^2$,

$$\frac{9}{a^2} - \frac{13}{4a^2} = 1 \Rightarrow \frac{36 - 13}{4a^2} = 1 \Rightarrow \frac{23}{4a^2} = 1.$$

Thus

$$a^2 = \frac{23}{4}, \quad b^2 = 23.$$

Therefore the standard equation is

$$\boxed{\frac{x^2}{23/4} - \frac{y^2}{23} = 1}.$$

Equivalently,

$$\boxed{4x^2 - y^2 = 23}.$$

2. A hyperbola centered at the origin has foci $(\pm 5, 0)$. One asymptote has slope $\frac{4}{3}$. Find the vertices.

Since the foci are $(\pm 5, 0)$, the hyperbola is horizontal:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

with

$$c^2 = a^2 + b^2.$$

Here $c = 5$, so

$$a^2 + b^2 = 25.$$

The asymptotes are

$$y = \pm \frac{b}{a}x.$$

Given one asymptote has slope $\frac{4}{3}$, we have

$$\frac{b}{a} = \frac{4}{3}.$$

Let $a = 3k$, $b = 4k$. Then

$$a^2 + b^2 = 9k^2 + 16k^2 = 25k^2 = 25,$$

so $k = 1$. Hence

$$a = 3, \quad b = 4.$$

The vertices are $(\pm a, 0)$, so

$$\boxed{(\pm 3, 0)}.$$

3. A point P lies on a hyperbola with foci $(\pm 13, 0)$, and the absolute difference of its distances to the foci is 10. If P also lies on the line $y = x$, find P .

For a hyperbola with foci $(\pm c, 0)$, the constant difference of distances equals $2a$. We are told this difference is 10, so

$$2a = 10 \quad \Rightarrow \quad a = 5.$$

Since $c = 13$, we have

$$b^2 = c^2 - a^2 = 169 - 25 = 144.$$

Thus the hyperbola is

$$\frac{x^2}{25} - \frac{y^2}{144} = 1.$$

Because P lies on $y = x$, substitute $y = x$:

$$\frac{x^2}{25} - \frac{x^2}{144} = 1.$$

Compute:

$$x^2 \left(\frac{1}{25} - \frac{1}{144} \right) = 1 \quad \Rightarrow \quad x^2 \cdot \frac{144 - 25}{3600} = 1 \quad \Rightarrow \quad x^2 \cdot \frac{119}{3600} = 1.$$

So

$$x^2 = \frac{3600}{119}, \quad x = \pm \frac{60}{\sqrt{119}}.$$

Since $y = x$, we get

$$P = \left(\frac{60}{\sqrt{119}}, \frac{60}{\sqrt{119}} \right) \quad \text{or} \quad P = \left(-\frac{60}{\sqrt{119}}, -\frac{60}{\sqrt{119}} \right).$$

Therefore,

$$P = \left(\pm \frac{60}{\sqrt{119}}, \pm \frac{60}{\sqrt{119}} \right)$$

with matching signs.

4. Two hyperbolas are given by

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{and} \quad \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1.$$

If they intersect, determine the coordinates of all intersection points in terms of a and b .

Let

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{1}$$

and

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1. \tag{2}$$

Add (1) and (2):

$$x^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) + y^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = 2.$$

Thus

$$(x^2 + y^2) \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = 2. \quad (3)$$

Subtract (2) from (1):

$$x^2 \left(\frac{1}{a^2} + \frac{1}{b^2} \right) - y^2 \left(\frac{1}{a^2} + \frac{1}{b^2} \right) = 0.$$

Since $\frac{1}{a^2} + \frac{1}{b^2} > 0$, this gives

$$x^2 - y^2 = 0 \quad \Rightarrow \quad y = \pm x.$$

So at any intersection point,

$$x^2 = y^2.$$

Substitute $y^2 = x^2$ into (1):

$$x^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = 1.$$

Hence

$$x^2 = \frac{1}{\frac{1}{a^2} - \frac{1}{b^2}} = \frac{a^2 b^2}{b^2 - a^2}.$$

Therefore intersection points exist only if $b^2 > a^2$, i.e. $b > a$.

Then

$$x = \pm \frac{ab}{\sqrt{b^2 - a^2}}, \quad y = \pm x.$$

So all intersection points are

$$\left(\frac{ab}{\sqrt{b^2 - a^2}}, \frac{ab}{\sqrt{b^2 - a^2}} \right), \left(\frac{ab}{\sqrt{b^2 - a^2}}, -\frac{ab}{\sqrt{b^2 - a^2}} \right), \left(-\frac{ab}{\sqrt{b^2 - a^2}}, \frac{ab}{\sqrt{b^2 - a^2}} \right), \left(-\frac{ab}{\sqrt{b^2 - a^2}}, -\frac{ab}{\sqrt{b^2 - a^2}} \right)$$

provided $b > a$. If $b \leq a$, there are no real intersections.

5. A hyperbola centered at the origin has vertices $(\pm 4, 0)$, and one of its asymptotes is perpendicular to the line $3x - 4y = 0$. Find the equation of the hyperbola.

Since the vertices are $(\pm 4, 0)$, the hyperbola is horizontal:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

with

$$a = 4, \quad a^2 = 16.$$

The line $3x - 4y = 0$ has slope $\frac{3}{4}$. A perpendicular line has slope

$$-\frac{4}{3}.$$

For the hyperbola, asymptotes are

$$y = \pm \frac{b}{a}x.$$

So one asymptote having slope $-\frac{4}{3}$ means

$$\frac{b}{a} = \frac{4}{3}.$$

With $a = 4$, this gives

$$b = \frac{4}{3} \cdot 4 = \frac{16}{3}, \quad b^2 = \frac{256}{9}.$$

Hence the equation is

$$\boxed{\frac{x^2}{16} - \frac{y^2}{256/9} = 1}.$$

Equivalently,

$$\boxed{16x^2 - 9y^2 = 256}.$$

6. Among all hyperbolas centered at the origin with asymptotes $y = \pm x$, one passes through $(5, 3)$ and another through $(13, 12)$. Without graphing, decide which hyperbola is closer to its center at the vertices, and by how much.

A hyperbola centered at the origin with asymptotes $y = \pm x$ must satisfy $b = a$. Hence its equation is either

$$\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1 \quad \text{or} \quad \frac{y^2}{a^2} - \frac{x^2}{a^2} = 1,$$

that is,

$$x^2 - y^2 = a^2 \quad \text{or} \quad y^2 - x^2 = a^2.$$

The distance from the center to a vertex is a .

Hyperbola through $(5, 3)$

Since $5^2 - 3^2 = 25 - 9 = 16 > 0$, the relevant equation is

$$x^2 - y^2 = a^2,$$

so

$$a^2 = 16 \quad \Rightarrow \quad a = 4.$$

Hyperbola through $(13, 12)$

Similarly,

$$13^2 - 12^2 = (13 - 12)(13 + 12) = 25,$$

so

$$a^2 = 25 \quad \Rightarrow \quad a = 5.$$

Thus the first hyperbola has vertices at distance 4 from the center, while the second has vertices at distance 5.

Therefore the hyperbola through $(5, 3)$ is closer to its center, by

